

Three-Week Research Plan

A Hybrid Quantum–Classical Materials Informatics Framework for
Hydrogen-Defect-Controlled Transport in Model Quantum Materials

Prepared for Dr. Başak Ekinici

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Abstract

This document presents a three-week project plan for developing a proof-of-concept hybrid quantum–classical materials informatics framework for studying hydrogen-defect-controlled transport descriptors in model quantum materials. The project models hydrogen-induced defects as local perturbations in lattice Hamiltonians, generates simulation-based datasets, extracts physics-informed descriptors, trains machine learning models, optimizes defect configurations, and validates selected small Hamiltonians using Qiskit-compatible quantum simulation. The purpose is not to claim the discovery of a real industrial material, but to build a scientifically safe, reproducible, and portfolio-ready computational framework connecting defect physics, machine learning, optimization, and quantum simulation.

English Correction

The original request can be written more naturally as:

“Now, can you please write everything in the same format as this?”

A clearer version for the current task is:

“Please convert the full project plan into the same structured Overleaf-ready format, including the scientific motivation, timeline, tasks, outputs, and safe claims.”

1 Final Project Direction

1.1 Main Idea

The project studies how hydrogen-induced defects may modify transport-related properties in a simplified model material. The main motivation is that modern computing devices face serious heat-management and transport-related challenges. Controlled defects may provide a possible route for tuning electronic or thermal transport descriptors in materials.

This project develops a small but meaningful materials informatics framework using:

- physics-based lattice simulations,
- synthetic defect datasets,
- machine learning prediction,
- feature-importance analysis,
- optimization,
- Qiskit-compatible quantum simulation.

1.2 Best Project Title

A Hybrid Quantum–Classical Materials Informatics Framework for Hydrogen-Defect-Controlled Transport in Model Quantum Materials

1.3 Simpler Project Title

Hydrogen-Defect Engineering for Transport Control in Model Materials Using Machine Learning and Qiskit

2 Scientific Model

The project begins with a simplified tight-binding Hamiltonian:

$$H = H_0 + H_{\text{defect}}. \quad (1)$$

The clean Hamiltonian is:

$$H_0 = -t \sum_{\langle i,j \rangle} c_i^\dagger c_j, \quad (2)$$

and the defect Hamiltonian is:

$$H_{\text{defect}} = \sum_i V_i c_i^\dagger c_i. \quad (3)$$

Here:

Symbol	Meaning
t	Hopping parameter controlling electron movement between neighboring sites.
V_i	Hydrogen-defect potential at lattice site i .
i	Lattice site index.
$c_i^\dagger$	Creation operator at site i .
c_i	Annihilation operator at site i .
H_0	Clean lattice Hamiltonian.
H_{defect}	Defect-induced perturbation Hamiltonian.

In the simplified matrix version used in the first stage, hydrogen defects are modeled as local potential changes:

$$\epsilon_i \rightarrow \epsilon_i + V_H. \quad (4)$$

3 What the Project Will Compute

The computational pipeline will calculate:

1. energy spectrum,
2. energy gap,
3. localization index,
4. defect density,

5. defect clustering,
6. transport descriptor,
7. machine learning prediction,
8. feature importance,
9. optimized defect configuration,
10. small Qiskit validation.

4 Scientific Claims

4.1 Safe Claim

The strongest and safest claim is:

This study develops a proof-of-concept computational framework for studying how hydrogen-induced defects may tune transport descriptors in model quantum materials.

Another safe version is:

We developed a physics-informed materials informatics pipeline and validated selected small Hamiltonians using Qiskit-compatible quantum simulation tools.

4.2 Claims We Should Avoid

The project should **not** claim:

- that we designed a real industrial material,
- that we solved the heating problem in computers,
- that IBM Quantum produced a real thermal material,
- that the transport descriptor is exact thermal conductivity,
- that the toy model directly represents a real material without further validation.

4.3 Safer Scientific Language

Use:

- “transport descriptor”,
- “thermal-transport proxy”,
- “transport-related score”,
- “model material framework”,
- “proof-of-concept computational study”,
- “Qiskit-compatible validation”.

Avoid:

- “exact thermal conductivity”,
- “real material design”,
- “industrial material discovery”,
- “IBM designed material”.

5 Best Target Output After Three Weeks

At the end of three weeks, the realistic target outputs are:

- working Python codebase,
- synthetic/simulation-based defect dataset,
- machine learning results,
- feature-importance plots,
- optimization results,
- small Qiskit simulation,
- first full manuscript draft,
- GitHub repository,
- short LinkedIn or portfolio explanation.

6 Overall Timeline

A three-week timeline is recommended. A two-week version may be possible, but it would be stressful and scientifically weaker. Three weeks gives a better chance of producing a manuscript-ready and portfolio-ready project.

- **Week 1:** Physics model and dataset generation.
- **Week 2:** Machine learning and optimization.
- **Week 3:** Qiskit validation and manuscript.

7 Week 1: Physics Model and Dataset Generation

7.1 Goal of Week 1

By the end of Week 1, the project should include:

- a simple lattice model,
- hydrogen defect configurations,
- energy spectrum calculation,
- localization index,
- transport descriptor,
- first dataset.

7.2 Day 1: Define the Project and Prepare Environment

Tasks

Create the project folder:

```
hydrogen_defect_transport/  
|  
|-- data/  
|-- notebooks/  
|-- src/  
|-- results/  
|-- figures/  
|-- manuscript/  
|-- README.md
```

Install or use:

```
numpy  
pandas  
scipy  
matplotlib  
scikit-learn  
xgboost  
optuna  
qiskit  
qiskit-aer  
qiskit-ibm-runtime
```

Create the first notebook:

`notebooks/01_lattice_model.ipynb`

Scientific Task

Write one paragraph defining the project:

We investigate hydrogen-induced defects as local perturbations in a model lattice Hamiltonian and study their effect on electronic localization and transport-related descriptors.

Output of Day 1

- project folder ready,
- environment ready,
- project paragraph written,
- first notebook created.

7.3 Day 2: Build the Clean Lattice Hamiltonian

Task

Start with a one-dimensional chain. For example:

$$N = 20, \quad t = 1.0. \quad (5)$$

Build the clean Hamiltonian:

$$H_0 = \begin{bmatrix} 0 & -t & 0 & \dots \\ -t & 0 & -t & \dots \\ 0 & -t & 0 & \dots \\ \dots & \dots & \dots & \dots \end{bmatrix}. \quad (6)$$

Calculate:

- eigenvalues,
- eigenvectors,
- energy spectrum.

Plot

Plot the eigenvalue spectrum for the clean lattice.

Output of Day 2

- clean lattice Hamiltonian,
- eigenvalue spectrum,
- first plot.

7.4 Day 3: Add Hydrogen Defects

Task

Add hydrogen defects as on-site potentials:

$$H = H_0 + \sum_i V_i |i\rangle \langle i|. \quad (7)$$

Choose:

- `defect_density` = 0.1,
- `defect_potential` = 0.5, 1.0, 2.0.

Test:

- no defect,
- one defect,
- random defects,
- clustered defects.

Plot

Compare spectra for:

- clean system,
- weak defect,
- strong defect,
- clustered defects.

Output of Day 3

- hydrogen-defect model working,
- comparison plots.

7.5 Day 4: Calculate Localization Index

Task

Use the inverse participation ratio:

$$IPR = \sum_i |\psi_i|^4. \quad (8)$$

Interpretation:

- low IPR: delocalized state,
- high IPR: localized state.

Calculate average IPR for each defect configuration.

Plots

- defect potential versus localization index,
- defect density versus localization index.

Output of Day 4

- localization descriptor,
- localization plots.

7.6 Day 5: Define Transport Descriptor

A simple but defensible descriptor is required.

For the first version:

$$T_{\text{score}} = \frac{1}{1 + \alpha \cdot IPR + \beta \cdot D}, \quad (9)$$

where:

Symbol	Meaning
IPR	Localization index.
D	Defect density.
α, β	Weighting constants.

This should be called a:

- transport descriptor,
- thermal-transport proxy,
- transport-related score.

It should **not** be called exact thermal conductivity.

Task

Calculate:

- transport descriptor,
- energy gap,
- average localization.

Plots

- defect density versus transport descriptor,
- localization versus transport descriptor,
- defect potential versus transport descriptor.

Output of Day 5

- first complete physical descriptor pipeline.

7.7 Day 6: Generate the Dataset

Task

Generate 1000–3000 samples.

Each sample should include:

- defect density,
- defect potential,
- number of defects,
- defect pattern,
- mean defect distance,
- clustering index,
- energy gap,
- mean IPR,
- transport descriptor.

Save:

`data/hydrogen_defect_dataset.csv`

Output of Day 6

- dataset created,
- CSV saved,
- basic summary statistics.

7.8 Day 7: Week 1 Review and Figures

Task

Clean all plots.

Figures for the manuscript:

- clean versus defect lattice diagram,
- energy spectrum comparison,
- defect potential versus localization,
- defect density versus transport descriptor.

Writing

Start the Methods section:

- lattice model,
- defect modeling,
- descriptor calculation.

Output of Day 7

- Week 1 results ready,
- first Methods draft.

8 Week 2: Machine Learning and Optimization

8.1 Goal of Week 2

By the end of Week 2, the project should include:

- machine learning model,
- prediction results,
- feature importance,
- optimized defect configuration,
- figures and tables.

8.2 Day 8: Prepare ML Dataset

Task

Use the following features:

- defect density,
- defect potential,
- number of defects,
- mean defect distance,

- clustering index,
- energy gap,
- mean IPR.

Target:

`transport_descriptor`

Use:

train/test split = 80/20. (10)

Models:

- Random Forest Regressor,
- XGBoost Regressor,
- Gradient Boosting Regressor.

Metrics:

- R^2 ,
- RMSE,
- MAE.

Output of Day 8

- ML-ready dataset,
- baseline regression results.

8.3 Day 9: Train Main ML Models

Train:

- Random Forest,
- XGBoost,
- Gradient Boosting.

Compare results using the following table:

Model	MAE	RMSE	R^2
Random Forest			
XGBoost			
Gradient Boosting			

Output of Day 9

- best ML model selected,
- results table.

8.4 Day 10: Feature Importance

Task

Use:

- permutation importance,
- Random Forest feature importance,
- SHAP if time allows.

Find which defect descriptors control the transport descriptor most strongly.

Expected important features:

- defect density,
- defect potential,
- localization index,
- clustering index.

Plot

Create a feature importance bar plot.

Output of Day 10

- interpretability result,
- feature importance figure.

8.5 Day 11: Optimization with Optuna

Task

Use Optuna to search for defect configurations that maximize the transport descriptor:

$$\max T_{\text{score}}. \quad (11)$$

Input variables:

- defect density,
- defect potential,
- clustering index.

Output:

- best defect configuration.

Interpretation

If a high transport descriptor indicates better heat spreading in this model, then the optimized configuration suggests:

Lower localization and controlled defect density may support better transport behavior.

Output of Day 11

- optimized defect configuration,
- optimization plot.

8.6 Day 12: Add Comparison Cases

Compare four cases:

1. no defect,
2. weak random hydrogen defects,
3. strong random hydrogen defects,
4. clustered hydrogen defects.

For each case report:

- energy gap,
- mean IPR,
- transport descriptor.

Case	Energy Gap	Mean IPR	Transport tor	Descrip-
Clean				
Weak random defect				
Strong random defect				
Clustered defect				

Output of Day 12

- strong Results section structure.

8.7 Day 13: Prepare Manuscript Figures

Required figures:

1. Figure 1: Project workflow,
2. Figure 2: Lattice plus hydrogen defect model,
3. Figure 3: Energy spectrum comparison,
4. Figure 4: Localization/IPR result,
5. Figure 5: ML prediction actual versus predicted,
6. Figure 6: Feature importance,
7. Figure 7: Optimization result,
8. Figure 8: Qiskit validation.

Output of Day 13

- almost all classical/ML figures ready.

8.8 Day 14: Week 2 Review and Writing

Draft these sections:

- Introduction skeleton,
- Materials and Methods,
- ML Pipeline,
- Results: classical simulation,
- Results: ML prediction.

Output of Day 14

- manuscript 40–50% complete,
- all ML results ready.

9 Week 3: Qiskit Validation and Manuscript

9.1 Goal of Week 3

By the end of Week 3:

- small Hamiltonian mapped to qubits,
- Qiskit simulation completed,
- optional IBM hardware run attempted,
- manuscript draft completed.

9.2 Day 15: Build Small Hamiltonian for Qiskit

Task

Choose a very small system:

$$N = 2 \quad \text{or} \quad N = 4. \tag{12}$$

Create a simple qubit Hamiltonian using Pauli operators:

$$H = aZ_0 + bZ_1 + cX_0X_1 + dY_0Y_1. \tag{13}$$

A defect can be represented as a modified on-site term:

$$V_{\text{defect}}Z_i. \tag{14}$$

Output of Day 15

- small qubit Hamiltonian,
- exact classical eigenvalue.

9.3 Day 16: Qiskit Simulator Run

Task

Use Qiskit Aer or Estimator.

Calculate:

- expectation value, or
- approximate ground-state energy.

Compare:

Method	Ground-State Energy
Classical diagonalization	
Qiskit simulator	

Output of Day 16

- Qiskit simulator result.

9.4 Day 17: VQE or Estimator Version

Task

If time allows, implement VQE.

Use:

- simple ansatz,
- COBYLA or SPSA optimizer,
- small qubit Hamiltonian.

Compare:

- exact result,
- VQE result.

Output of Day 17

- Qiskit validation figure or table.

9.5 Day 18: IBM Quantum Hardware Run

Task

Try running the smallest circuit on IBM Quantum. This is optional but valuable.

If hardware queue or account issues happen, do not stop the project. Simulator results are enough for the proof-of-concept version.

If Hardware Works

Use this sentence:

Selected small defect Hamiltonians were evaluated using Qiskit Runtime on IBM Quantum hardware.

If Hardware Does Not Work

Use this sentence:

Selected small defect Hamiltonians were evaluated using Qiskit Aer and IBM Quantum-compatible workflows.

Output of Day 18

- IBM/Qiskit section ready.

9.6 Day 19: Write Full Manuscript

Manuscript Structure

1. Title,
2. Abstract,
3. Introduction,
4. Related Work,
5. Methods,
 - (a) Lattice model,
 - (b) Hydrogen defect modeling,
 - (c) Transport descriptor,
 - (d) Materials informatics dataset,
 - (e) Machine learning,
 - (f) Qiskit validation,
6. Results,
7. Discussion,
8. Limitations,
9. Conclusion.

Output of Day 19

- first full manuscript draft.

9.7 Day 20: Improve Scientific Language

Task

Make the claims safer.

Replace:

thermal conductivity

with:

thermal transport descriptor, transport proxy, transport-related score.

Replace:

real material design

with:

model material framework, proof-of-concept computational study.

Replace:

IBM designed material

with:

selected Hamiltonians were evaluated using Qiskit/IBM Quantum-compatible tools.

Output of Day 20

- scientifically safer manuscript.

9.8 Day 21: GitHub and Final Review

GitHub Repository Structure

```
hydrogen-defect-transport/  
|  
|-- README.md  
|-- requirements.txt  
|-- data/  
|   |-- hydrogen_defect_dataset.csv  
|-- notebooks/  
|   |-- 01_lattice_model.ipynb  
|   |-- 02_dataset_generation.ipynb  
|   |-- 03_ml_prediction.ipynb  
|   |-- 04_optimization.ipynb  
|   |-- 05_qiskit_validation.ipynb  
|-- src/  
|   |-- lattice.py  
|   |-- defects.py  
|   |-- descriptors.py  
|   |-- ml_models.py  
|   |-- qiskit_validation.py  
|-- figures/  
|-- manuscript/
```

Final Output

- GitHub-ready code,
- manuscript draft,
- figures,
- results table,
- portfolio summary.

10 Daily Working Schedule

Since the project requires focused work, the suggested daily structure is:

Morning Block: 09:00–12:00

Main coding or mathematical modeling.

No reading. No YouTube. Only implementation.

Lunch Break: 12:00–13:00

Rest.

Afternoon Block: 13:00–16:00

Experiments, plots, debugging, and ML results.

Short Break: 16:00–16:30

Rest.

Evening Block: 16:30–18:30

Writing manuscript.

Night Light Block: 20:30–21:30

Read one related paper or tutorial.

Final 15 Minutes

Write a short daily log:

```
Today I completed:
Problem I faced:
Tomorrow I will do:
One sentence for the paper:
```

This daily log will help write the manuscript quickly.

11 Why the First Three Days Are Critical

The first three days are critical because they establish whether the project is technically moving.

Day	Critical Goal
Day 1	Project folder, environment, and project definition.
Day 2	Clean one-dimensional lattice Hamiltonian.
Day 3	Hydrogen defect potential added and plotted.

After Day 3, we will know whether the project is technically stable.

12 Minimum Publishable Version

If time becomes tight, reduce the scope to:

- one-dimensional lattice only,
- 1000 samples,

- Random Forest and XGBoost,
- permutation importance,
- Optuna,
- Qiskit simulator only,
- no real IBM hardware.

This is still sufficient for:

proof-of-concept manuscript + GitHub portfolio.

13 Better Version

If everything goes well, extend to:

- one-dimensional plus small two-dimensional lattice,
- 3000–5000 samples,
- SHAP analysis,
- Optuna optimization,
- Qiskit VQE,
- optional IBM hardware run,
- more polished manuscript.

14 Clean Final Project Story

The final project story should be:

Heat management is a major issue in modern computing. Defect engineering is a possible route to tune transport properties in materials. Here, we model hydrogen-induced defects as local perturbations in lattice Hamiltonians. We generate a simulation-based materials informatics dataset, extract physics-informed descriptors, train machine learning models to predict transport behavior, identify important defect features, optimize defect configurations, and validate selected small Hamiltonians using Qiskit-compatible quantum simulation.

15 Honest Assessment

A strong first version is realistic in three weeks if the work is focused. The safest target is not journal submission within three weeks. The safest target is:

complete manuscript draft + complete code + complete figures + GitHub portfolio in three weeks.

After that, one additional week can be used to polish the work for journal submission.

16 Recommended References

16.1 Tight-Binding and Lattice Hamiltonians

1. David Tong, *Applications of Quantum Mechanics*. Tight-binding discussion.
<https://www.damtp.cam.ac.uk/user/tong/aqm/aqmtwo.pdf>
2. Tight-binding model notes.
<https://ssp.utasphys.cloud.edu.au/3-1d/3-3-tightbinding/>
3. Kwant tight-binding introduction.
<https://kwant-project.org/doc/latest/tutorial/introduction>

16.2 Eigenvalue Calculations

1. SciPy `eigh` documentation.
<https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.eigh.html>
2. NumPy `eigh` documentation.
<https://numpy.org/doc/stable/reference/generated/numpy.linalg.eigh.html>

16.3 Localization and Defect Physics

1. Inverse Participation Ratio overview.
<https://www.emergentmind.com/topics/inverse-participation-ratio-ipr>
2. Anderson localization background notes.
<https://web.mit.edu/8.334/www/grades/projects/projects19/GuanChenguang.pdf>

16.4 Machine Learning and Optimization

1. Scikit-learn documentation.
<https://scikit-learn.org/stable/>
2. XGBoost documentation.
<https://xgboost.readthedocs.io/>
3. Optuna documentation.
<https://optuna.org/>

16.5 Qiskit and Quantum Machine Learning

1. Qiskit documentation.
<https://docs.quantum.ibm.com/>
2. Qiskit Machine Learning quantum kernel tutorial.
https://qiskit-community.github.io/qiskit-machine-learning/tutorials/03_quantum_kernel.html
3. IBM Quantum kernel training tutorial.
<https://quantum.cloud.ibm.com/docs/tutorials/quantum-kernel-training>